Food spoilage is considered as a complex phenomenon where by a combination of microbial and bio-chemical activities take place. Due to such activities various types of metabolites are formed which aid in spoilage (Figure 5.1).



Figure 5.1: Food spoilage

The ease with which foods are spoiled depends upon factors described. The foods are thus, divided into different classes. They are:

i. Perishable foods

These foods are readily spoiled; require special preservation and storage condition for use. This includes, foods such as dairy products, eggs, poultry, meat, fish, fruits and vegetable. These foods get spoiled easily by natural enzymes.

ii. Semi – perishable foods

This class of foods if properly stored can be used for a longer duration. These foods include processed cereals, pulses and their products like flour, semolina, parched rice and popcorn. Shelf life of these products depends on the storage temperature and moisture in the air. Foods like potato, onion, nuts, frozen foods and certain canned foods can be stored for a week to a couple of months at room temperature without any undesirable changes in the products.

iii. Non – perishable foods

These foods remain stable for long period unless handled improperly. Non-perishable foods include sugar, jaggery, hydrogenated fat, vegetable oil, ghee, whole grains, dhals, whole nuts and processed foods like dry salted fish/meat, papads, canned foods, jams and murabbas. These foods do not spoil unless they are handled carelessly.

5.2.1 Causes of Food Spoilage

Food and water may be infected by germs. When flies carrying germs sit on our food, they pass on the germs to our food. There are various factors which are responsible for food spoilage such as.

- Microorganism
- Insects
- Rough handling
- Transport
- improper storage
- enzyme activity (Chemical reaction)
- unhygienic conditions

physical changes, such as those caused by freezing, burning, drying pressure.

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